ELICIT Architecture: Technical Specification

Abstract

ELICIT is a multi-agent simulation for studying institutional cooperation. Language-model agents repeatedly choose institutions, contribute resources, sanction peers, absorb climate shocks, and revise social beliefs. The architecture links game mechanics, memory, reputation, democratic rule change, and data export in one repeated pipeline.

1 Overview

ELICIT studies how cooperative institutions appear, persist, change, or fail. It places language-model agents inside a repeated public goods game. The game includes social pressure, reputation, democratic governance, climate shocks, and loss-and-damage transfers.

Technically, the system is organized as a repeated round loop. Each round begins from shared rules and agent state. It then executes institution choice, contribution, sanctioning, climate accounting, payoff updates, and post-round cognition. The updated state becomes the starting point for the next round.

- The outer layer runs many configured simulation runs.
- The shared rule state stores all live parameters.
- The agent layer holds wealth, memory, beliefs, and reputation.
- The institutional layer computes contributions, sanctions, and rewards.
- The climate layer computes damage and redistribution.
- The output layer records full round and agent histories.

What this means in practice: The platform is not a static game. It is a closed feedback system where actions change wealth, memory, reputation, and sometimes the rules themselves.

1.1 Single-round pipeline

The figure summarizes the main data flow across one round. It avoids implementation names and shows only the architectural roles. Solid arrows show ordinary round flow. Dashed arrows show feedback into later rounds or periodic governance.

What this means in practice: A reader can follow the system without knowing the code. The round converts rules and agent state into actions, outcomes, records, and updated social context.

2 Scenario State and Shared Rules

The shared rule state contains the parameters that define a run. These include agent counts, round counts, public goods multipliers, punishment and reward ratios, subsidy settings, climate shock settings, loss-and-damage policy weights, voting intervals, and model settings. The same state is read by the game, the climate mechanism, and the governance process.

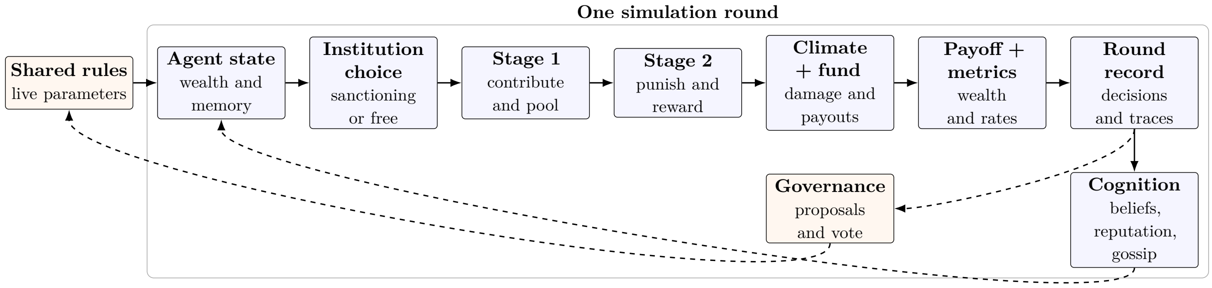

Figure 1: Single-round architecture of the ELICIT simulation. Dashed arrows show feedback into later rounds.

The important design choice is mutability. Most parameters begin as configuration values. Some can later be changed by democratic voting. This makes the rule state both an initial setup and a live institutional constitution.

- Static values define the initial experiment.
- Mutable values allow rule change during the run.
- Governance changes apply from the next round onward.

What this means in practice: Rules are not only inputs. They can become outcomes of collective decision-making.

3 Agent Design

Agents differ by economic group and climate exposure. Developed-nation agents begin with higher wealth, higher contribution capacity, lower vulnerability, and higher historical emissions. Developing-nation agents begin with lower wealth, lower capacity, and higher vulnerability. In the climate scenario, starting endowments can be tied to nominal GDP values of real countries.

Each agent also carries a small internal state. This state includes current wealth, cumulative payoff, group identity, recent episodic memory, a belief scratchpad, reputation, and parser-failure counts. The memory and belief fields are updated after each round. They are then included in the next prompt.

- Material state records wealth, payoff, and contribution capacity.
- Social state records trust, reputation, gossip, and peer assessments.
- Climate state records vulnerability, damage, payments, and transfers.

What this means in practice: Agents are not interchangeable players. They differ in capacity, exposure, memory, and social standing.

The platform can also run non-language-model baselines. Random agents provide stochastic behavior. Greedy agents represent short-horizon defection. Mixed populations allow language-model agents to interact with these simpler types.

What this means in practice: Baselines make the main results easier to interpret. They show whether language-model agents behave differently from simple heuristics.

4 Language-Model Interface

Each language-model agent receives structured prompts before major decisions. The prompt includes the agent's wealth, group type, cumulative payoff, live rules, recent outcomes, belief state, peer information, and persona framing. For institution choice, the prompt also includes a compact decision card that explains the payoff logic and risk profile of each institution.

The system uses a locally hosted model endpoint through an OpenAI-compatible interface. It supports ordinary chat-style requests and a separate raw path for reasoning-oriented models. It requests structured output when possible. If the first answer does not match the expected schema, a repair retry is attempted. Failures are counted and saved.

- Prompts combine economic, social, and institutional context.
- Responses are parsed into structured decisions.
- Invalid responses are repaired when possible.
- Parsing errors remain visible in the simulation record.

What this means in practice: The architecture treats model output as useful but unreliable. It validates decisions instead of assuming they are well-formed.

5 Round Execution

5.1 Institution choice

At the start of a round, agents select an institution. The two options are a sanctioning institution and a sanction-free institution. The sanctioning institution allows later punishment and reward. The sanction-free institution only runs the contribution stage.

In climate mode, this choice can be fixed by group. Developed agents are routed to the binding sanctioning institution. Developing agents are routed to the non-binding sanction-free institution. This mirrors an asymmetric treaty setting where developed nations carry stronger enforcement obligations.

What this means in practice: Institution choice decides the rule environment for the round. In climate mode, the asymmetry is imposed by scenario design rather than by free choice.

5.2 Stage 1: public goods contribution

Within each institution, members choose an integer contribution. The contribution is bounded by either a fixed per-round endowment or by accumulated wealth in climate-budget mode. The institution sums all contributions, applies the public goods multiplier, and distributes the resulting return equally among members.

- Higher total contribution increases the public pool.
- Equal distribution creates a free-rider incentive.
- The cooperation rate is measured against each agent's contribution cap.

What this means in practice: Stage 1 creates the central tension. The group benefits from contribution, but each individual can gain by holding back.

5.3 Stage 2: punishment, reward, and subsidy

Only the sanctioning institution runs Stage 2. Each member receives a list of co-members ranked by contribution relative to the round average. Below-average contributors can be marked as free riders. Trust scores from prior social evaluation can also be included.

Agents allocate punishment and reward tokens within a budget. The parser maps anonymous labels back to actual agents. It can block punishments against above-average contributors under a rule-of-law constraint. If allocations exceed the budget, they are scaled down proportionally.

The final Stage 2 payoff depends on unspent budget, received rewards, and received punishments. A subsidy mechanism can also pool part of punishment costs and redistribute it to top contributors.

What this means in practice: Stage 2 adds social enforcement. It can support cooperation, but it also creates costs and a second-order free-rider problem.

6 Climate Risk and Loss-and-Damage Accounting

Climate shocks can occur by stochastic probability or by a fixed schedule. When a shock occurs, each agent's gross damage is computed from a base damage value, shock severity, and individual vulnerability. Developing agents therefore face higher expected damage when their vulnerability values are higher.

The loss-and-damage fund adds redistribution. It collects contributions according to its replenishment policy. On shock rounds, it pays only developing agents. Payouts are weighted by damage and can be adjusted toward poorer agents through an equity weight. Coverage is capped by a maximum fraction of gross damage and by the fund balance. Unused funds roll forward.

- Gross damage reduces wealth.
- Fund payouts offset part of that damage.
- Net climate transfer equals payout minus damage.
- Fund contributions, balances, damages, and payouts are recorded each round.

What this means in practice: Climate shocks turn cooperation into a survival problem. Redistribution lets the model test whether institutions protect vulnerable agents or only reward wealthy ones.

7 Cognitive Feedback

After payoffs are computed, each agent receives a feedback object for the round. This updates the agent's short episodic memory. If belief tracking is active, the agent also produces a compact belief update. That update contains trust classifications, a revised institutional strategy, and general observations.

This design replaces a long raw memory window with a compressed working model. The agent does not need to reread the whole history. It carries forward a short interpretation of what the last round implied.

What this means in practice: Memory is semantic rather than archival. Agents remember what they think matters, not every raw event.

A separate reputation process compares each agent's stated reasoning with its actual contribution. Other agents score the consistency of that behavior. The incoming scores are averaged into a reputation value. Low-trust cases can be turned into gossip bulletins for the next round.

- Reputation links words to actions.
- Gossip spreads selected low-trust evidence.

- Bulletins are limited to avoid prompt bloat.
- The next round prompt includes this social evidence.

What this means in practice: The social layer makes behavior persistent. A misleading or selfish action can affect later treatment through reputation and gossip.

8 Democratic Governance

At a configured interval, agents enter a governance session. Each agent proposes one rule change from a whitelist of numeric parameters. The whitelist covers sanction strength, reward strength, Stage 2 budget, punishment caps, subsidy settings, and selected loss-and-damage weights in climate mode. Proposals are clamped to a safe range and duplicate rule-value combinations are removed.

Before voting, recent round data are summarized into simple predictions. The prediction step uses recent cooperation and free-rider rates to annotate proposals with estimated welfare effects. Agents then vote on the annotated proposals. A winning proposal is applied to the live rule state and remains active unless changed later.

- Proposal options are randomized to reduce ordering bias.
- Each agent casts one vote.
- Ties are broken randomly.
- The full voting record is saved with the round snapshot.

What this means in practice: Governance makes institutional design endogenous. Agents do not only play under rules; they can revise the rules after seeing outcomes.

9 Data Pipeline and Analysis Outputs

Every round produces a structured snapshot. The snapshot includes institution choices, contributions, sanction decisions, reasons, parser metadata, belief states, reputation values, climate damage, fund transfers, wealth, payoffs, and summary metrics. These snapshots are saved into timestamped JSON files. The file names encode model, batch label, seed, agent count, and round count.

A separate export process flattens the nested JSON into tidy round-level and agent-level tables. It also computes summary statistics such as final average payoff, mean reputation, parsing failure rate, contribution mean, contribution variation, and institution switching events.

The visualization layer produces diagnostic plots for selected runs. Plots cover contribution trajectories, institution membership, cumulative payoff rankings, punishment and reward flows, reputation evolution, and loss-and-damage fund dynamics. An interactive browser dashboard can load a JSON file by drag-and-drop. It displays macro dynamics, agent histories, sanction networks, fund analytics, and voting history.

What this means in practice: The data design preserves both aggregate trends and individual explanations. This lets the same run support statistical analysis and case-level inspection.

10 Experimental Conditions

The platform supports ablation studies. A control condition disables the cognitive modules. A reputation condition enables reputation and gossip without voting. A voting condition enables democracy without reputation and gossip. A full condition enables all major modules. Random and greedy baselines define behavioral reference points.

- Control isolates the base public goods mechanism.
- Reputation tests social monitoring and gossip.
- Voting tests institutional self-revision.
- Full combines cognition, reputation, gossip, and governance.
- Baselines show how simple non-cooperative agents behave.

What this means in practice: The ablation design helps assign causal weight. It separates the effects of incentives, social cognition, and rule change.

11 Interpretive Logic

The baseline analysis argues that defection is individually attractive without climate risk. Cooperation can still appear because language models often show a cooperative bias. That bias is not the same as incentive-compatible cooperation. It may sustain institutions even when self-interest points elsewhere.

Climate risk changes the interpretation. When shocks damage vulnerable agents and redistribution protects them, cooperation can become materially rational. The loss-and-damage mechanism is therefore not an add-on. It is central to testing whether treaty-style cooperation can become a survival strategy rather than a moral preference.

The design also exposes a fairness problem. Absolute contribution thresholds can punish low-capacity developing agents even when their proportional effort is high. This makes enforcement potentially regressive unless capacity and vulnerability are included in the rule design.

What this means in practice: The architecture is built to test a specific claim. Institutions must be judged not only by cooperation rates, but also by who pays, who is protected, and who gets punished.

12 Summary

ELICIT links repeated strategic interaction with memory, reputation, governance, climate risk, and redistribution. The core loop is simple: agents act, institutions compute outcomes, the environment records effects, and cognition updates future behavior. The distinctive feature is feedback. Material outcomes, social information, and rule changes all return to shape later decisions.

What this means in practice: The system is best understood as a laboratory for institutional feedback. It asks how cooperation changes when agents can remember, judge, punish, compensate, and vote.